

HIKET — discussion memo

Findings that may serve the Discussion, and are not yet strong enough for the storyline

Started 12 August 2026 — living document, revised continuously

What this is

A holding place. Analyses accumulate faster than the storyline can absorb them, and the ones that do not fit are either lost or, worse, quietly promoted without the caveat that kept them out. This document keeps them with their caveat attached.

Admission rule. An item belongs here if it could plausibly earn a sentence in the Discussion *and* there is a stated reason it cannot yet. If there is no such reason, it belongs in the manuscript. If there is no plausible sentence, it belongs in **doublechecks/** and nowhere else.

Status marks. ●●● measured and robust, one step from the storyline ●●○ measured but confounded, or resting on a single source ●○● suggestive; needs new data or a new test

Maintenance. Items are *deleted* when they graduate or die, leaving one line in the tables at the end. This document should not grow monotonically; if it does, it has become a diary rather than a memo.

1 Open items

●●○ **D1 — The 2006–2024 model–observation disagreement is confined to the mineral soil**
Claim. The six models do not fail to reproduce the recent SOC trend in general. They reproduce the *organic layer*, and disagree with the *mineral soil*.

Over 2006–2024, on the 429 plots measured in both campaigns. The model trajectories are *unweighted* plot means, so the unweighted column is the one that may be compared with them; the region-weighted column is the nationally representative figure and is shown so the two bases are never mixed.

compartment	unweighted	region-weighted
	Mg C ha ^{−1} yr ^{−1}	
organic layer	−0.094	−0.082
mineral 0–40 cm	+0.232	+0.181
whole measured profile	+0.138	+0.098
the six models (unweighted)	−0.02 to	−0.43

Basis warning, and it bites elsewhere. The observed trend quoted as +0.209 in `HIKET_next_session` and `M&M_parameterization_working_document` is the *unweighted* paired 2006–2024 rate. The region-weighted equivalent is +0.139, and LUKE’s official weighted figure is +0.111. All three are defensible; mixing them is not. Model comparisons must use unweighted; anything described as national must use weighted.

The organic layer (-0.094 unweighted) falls *inside* the model range. The entire discrepancy sits in the mineral soil — which is independently the least reliable compartment: samples are cut from the wall of a spade-dug pit rather than taken volumetrically (Kramarenko 2012, §4.4), proportional bias appears between *every* campaign pair in both mineral layers while the humus layer shows none, and $\sim 75\%$ of the first campaign’s mineral carbon concentrations were predicted from loss-on-ignition rather than measured.

Supporting arithmetic: the organic decline is also of the magnitude a falling litter input predicts. For a first-order pool a fractional input decline δ propagates as $\delta(1 - e^{-kt})$; with an organic-layer turnover of 30–40 yr over an 18-year window that factor is ≈ 0.4 , so the $\sim 13\%$ input decline in the Tupek series predicts $\sim 5\%$ stock loss against -6.6% observed. A 5-year-turnover pool would have predicted $\sim 13\%$, which is not what happened.

Why this is not in the storyline. (i) The organic layer is still a *layer*, defined by a field boundary, and that boundary moved: Δ_{organic} against $\Delta_{\text{mineral 0–20}}$ gives slope -0.25 ($r = -0.16$, $p = 0.0007$). The movement accounts for only $\sim 25\%$ of the organic loss, so $\sim 75\%$ survives — but the compartment is not boundary-proof. Only the whole-profile total is, and that shows $+0.20 \pm 0.11 \text{ kg m}^{-2}$, *not significant*. (ii) The organic total is the sum of a litter layer that rose 27% and a humus layer that fell 12%, and neither component behaves sensibly on its own (item D2). The aggregate may be right for the wrong reasons. (iii) The model range quoted is from the $\sigma = 0.72$ run and predates both the litter and the dating corrections to the target.

What would promote it. Re-running the comparison after the 1985 litter and sampling-year corrections land; and a model-side check that the models’ *own* organic pool, not just their total, matches. **What would kill it.** Evidence that the organic/mineral boundary moved much more than the -0.25 slope implies. *Code: doublechecks/organic_mineral_boundary.R. Added 2026-08-12.*

●●● D2 — The measured litter layer does not behave like a litter pool

Claim. The litter layer (LM) grew 27% between 2006 and 2024 while its input fell, and no plot-level mechanism explains it.

Aggregate: input -4.9% (2004–08 $\rightarrow$ 2020–24), litter layer $+27.7\%$, humus -11.8% . Per plot, neither candidate mechanism holds: growth tracking input gives $r = +0.09$ ($p = 0.07$), and reclassification out of the humus gives $r = -0.04$ ($p = 0.45$, slope -0.01). Cross-sectionally the litter layer has no relation to modelled litter input in either campaign ($r = +0.03$, $+0.01$).

The surviving explanation is a *uniform campaign-level* change in what was called litter in 2024: plot-level correlations are blind to an offset applied equally to every plot, which is exactly the signature observed.

The null cross-sectional correlation cannot separate “the litter layer is not behaving like a litter pool” from “the Tupek series carries little plot-level signal”. The latter is independently known (litter does not order SOC across plots; $R^2 \approx 0$). So the real-growth hypothesis is closer to *untestable with this predictor* than rejected. This must not be reported as evidence that the litter layer is spurious.

What would promote it. An independent plot-level litter predictor with demonstrated skill; or confirmation from LUKE that the 2024 litter sampling instruction differed. *Code: doublechecks/litter_layer_vs_inp. Added 2026-08-12.*

●●● D3 — Plot-level SOC change is not resolvable between campaigns

Claim. The 95% limits of agreement between any two campaigns are of the same size as the stock being measured.

Bland–Altman, whole profile: -30.5 to $+43.6$ Mg C ha $^{-1}$ for 1985–2006 and -39.0 to $+42.9$ for 2006–2024 — about 115% of the mean stock in both cases. For the mineral 20–40 cm layer it is $\sim 220\%$. This is a model-free statement: it uses only two measurements of the same plots.

It is the quantitative form of “the posteriors are overconfident”, and it needs no model to establish, which makes it the cleanest available support for treating plot-years as far less informative than their count suggests.

Placement, not strength, is what keeps this out of the storyline: it is an argument about the error model and the likelihood, and the storyline does not yet have a home for it. It may belong in the M&M working document instead of the Discussion.

Code: `Reporting/August2026_internal_report/build_bland_altman_campaigns.R`; figures there and manuscript Fig. S10(c). *Added 2026-08-12.*

2 Graduated

Items promoted into the manuscript. One line each; the content lives there now.

date	item	where it went
—	—	(none yet)

3 Retired

Items removed as falsified or superseded. Kept as one line so they are not rediscovered and re-argued.

date	item	what killed it
2026-08-12	1985 depth distribution is distorted	The per-plot decay rate puts 1985 ($\lambda = 0.038$) <i>between</i> 2006 (0.031) and 2024 (0.041). Profile shape does not single out 1985; the campaign difference is a level, not a shape.
2026-08-12	The subsoil offset is a stoniness-correction artefact	No relation to coarse fraction ($\rho = -0.08$, $p = 0.15$).
2026-08-12	1985 is noisier than the later campaigns	Subsoil repeatability $r = 0.63$ (1985–2006) vs 0.65 (2006–2024); proportional bias present in every pair. Whatever is wrong with 1985 is a level, not scatter.